

المحاضرة الخامسة

Operators in C++

- ❖ The table below lists all possible operators that can be executed in C++ programming.

#	Types of C operator	Symbols of operators
1	Arithmetic operators	+ → Addition - → Subtraction * → Multiplication / → Division % → Modulo (remainder after division)
2	Logical operators	&& → Logical AND → Logical OR ! → Logical NOT
3	Assignment operators	= → Assigns value of the left side to the right side += → a+=b is same as a = a+ b -= → a-=b is same as a = a- b *= → a*=b is same as a = a* b /= → a/=b is same as a = a/ b %= → a%=b is same as a = a% b

Operators in C++ (Cont...)

#	Types of C operator	Symbols of operators
4	Relational operators	< → Less than > → Greater than <= → Less than or equal to >= → Greater than or equal to == → is equal to != → is not equal to
5	Increment and Decrement operators	++ → increment value by 1 -- → decrement value by 1
6	Conditional operators	? → Example: x = (a>b)?a:b; <u>it means</u> If (a>b) x = a; else x = b;
7	Bitwise operators	& → bitwise AND → bitwise OR ^ → bitwise exclusive-OR << → Shift Left (it multiplies number by 2) >> → Shift Right (it divides number by 2)

Operators in C++ programming – Explanation

❖ Arithmetic operators are two types

➤ Unary operators: operators that operate on a single operator

Example

`b = a++;`

➤ Binary operators: operators that operate with two operators

Example

`c = a+b;`

- '+' is the operator known as addition operator,
- 'a' and 'b' are operands
- The addition operator tells the compiler to add both of the operands 'a' and 'b' and put them in c variable

Logical operators:

- The result of the operation of a logical operator is a **Boolean** value either *true* or *false*.
- used to test one or more condition or make decisions

Operators in C++ programming – Explanation

- ❖ Relational operators are used to compare two numbers and taking decisions based on their relation.
- ❖ Relational expressions are used in decision statements such as *if* , *for*.
- ❖ Conditional operator: It takes three arguments, *condition? exp1:exp2* if condition is true then execute exp1 otherwise exp2 will be executed.

❖ Bitwise Operators

- Bitwise operators are used to perform operation bit by bit.
- Bitwise operators may not be applied to float or double.

Assignment Operators in C++

- ❖ The Assignment operators in C++ are some of the C Programming Operator, which are useful to assign the values or the result of an expression to the declared variables.
- ❖ The equals (=) operator is the most commonly used assignment operator in C++.
- ❖ For example:

```
int i = 10;
```


Arithmetic operators – Example

```
#include <stdio.h>

int main() {
    int a = 9, b = 4, c;

    c = a+b;           //addition + → (a+b=13)
    printf("a+b = %d \n", c);

    c = a-b;           //subtraction - → (a-b=5)
    printf("a-b = %d \n", c);

    c = a*b;           //multiplication * → (a*b=36)
    printf("a*b = %d \n", c);

    c = a/b;           //division / → (a/b=2)
    printf("a/b = %d \n", c);

    c = a%b;           //modulo % → (a%b=1)
    printf("Remainder when a divided by b = %d \n", c);

    return 0;
}
```

Assignment Operators in C

- ❖ The below table displays all the assignment operators present in C Programming with an example.

C Assignment Operators	Example	Explanation
=	x = 25	Value 25 is assigned to x
+=	x += 25	This is same as → $x = x + 25$
-=	x -= 25	This is same as → $x = x - 25$
*=	x *= 25	This is same as → $x = x * 25$
/=	x /= 25	This is same as → $x = x / 25$
%=	x %= 25	This is same as → $x = x \% 25$

Assignment Operators in C++ – Example

- ❖ In following C++ assignment operators Program, two integer variables a and Total are used. And their values are 7 and 21, respectively. It shows the working functionality of all the Assignment Operators in C++ Programming Language.

```
/* Program for Assignment Operators in C*/  
#include <stdio.h>  
  
int main()  
{  
    int a = 7;  
    int Total = 21;  
  
    printf(" Value of the Total = %d \n", Total += a );  
    printf(" Value of the Total = %d \n", Total -= a );  
    printf(" Value of the Total = %d \n", Total *= a );  
    printf(" Value of the Total = %d \n", Total /= a );  
    printf(" Value of the Total = %d \n", Total %= a );  
  
    return 0;  
}
```

Assignment Operators in C++ – Example

```
1  /* Program for Assignment Operators in C */
2  #include <stdio.h>
3
4  int main()
5  {
6      int a = 7;
7      int Total = 21;
8
9      printf(" Value of the Total = %d \n", Total += a );
10     printf(" Value of the Total = %d \n", Total -= a );
11     printf(" Value of the Total = %d \n", Total *= a );
12     printf(" Value of the Total = %d \n", Total /= a );
13     printf(" Value of the Total = %d \n", Total %= a );
14
15     return 0;
16 }
17
```

output

```
Value of the Total = 28
Value of the Total = 21
Value of the Total = 147
Value of the Total = 21
Value of the Total = 0
```

Note:

- \n : means new line
- \t : means take a tab space then write

Assignment Operators in C++ – Example

- ❖ The ***printf*** statements will perform C++ Programming Assignment operations on **a** and **Total** and then display the output (result).
- ❖ Let us see the C Programming Operator functionality in this C Program

```
printf(" Value of the Total = %d \n ", Total += a );  
Total += a  
means → Total = Total + a = 21 + 7 = 28
```

```
printf(" Value of the Total = %d \n ", Total -= a );  
Total -= a  
means → Total = Total - a = 28 - 7 = 21
```

```
printf(" Value of the Total = %d \n ", Total *= a );  
Total *= a  
means → Total = Total * a = 21 * 7 = 147
```

Assignment Operators in C – Example

```
printf(" Value of the Total = %d \n ", Total /= a );
```

Total /= a

means $\rightarrow \text{Total} = \text{Total} / a = 147 / 7 = 21$

```
printf(" Value of the Total = %d \n ", Total %= a );
```

Total %= a

means $\rightarrow \text{Total} = \text{Total} \% a = 21 \% 7 = 0$ (the remainder of 21/7 is 0)

Increment and Decrement – Explanation

❖ Increment → ++

- **a++** is postfix, the expression is evaluated first and then the value is incremented.

Example

a = 10;

b = a++; // after this statement, a = 11, b = 10

- **++a** is prefix, the value is incremented first and then the expression is evaluated.

Example

a = 10;

b = ++a; // after this statement, a = 11, b = 11

Increment and Decrement – Explanation

❖ Decrement → -

- **a--** is postfix, the expression is evaluated first and then the value is decremented.

Example

```
a = 10;
```

```
b = a--; // after this statement, a = 9, b = 10
```

- **--a** is prefix, the value is decremented first and then the expression is evaluated.

Example

```
a = 10;
```

```
b = --a; // after this statement, a = 9, b = 9
```


Class #2 – Adding two double numbers

```
#include <stdio.h>
int main()
{
    double a, b, Sum = 0;

    printf("Enter the 1st number: ");
    scanf("%lf", &a);

    printf("Enter the 2nd number: ");
    scanf("%lf", &b);

    Sum = a + b;
    printf("%lf + %lf = %lf", a, b, Sum);

    return 0;
}
```

Output:

?



Build and run



```
Faris.Alghareb@Users-MacBook-Pro ~ % '/Users/user/Documents/'
Enter the 1st number: 5.5
Enter the 2nd number: 3.2
5.500000 + 3.200000 = 8.700000
Faris.Alghareb@Users-MacBook-Pro ~ %
```

Area calculation of a circle

Write a C program that reads the radius of a circle and prints its area and circumference.

Ans:

```
#include <stdio.h>
```

```
int main() {
```

?

```
return 0;  
}
```

THANK
YOU